

Multiple Choice (10 marks)

1. Which of the following statements about glucose metabolism is correct?
 - (a) The end-product of glycolysis in all cells is glucose
 - (b) Glycolysis doesn't require any enzymes
 - (c) In glycolysis, glucose is cleaved into two three-carbon products
 - (d) Glucose metabolism only occurs in the presence of oxygen
 - (e) In glycolysis, glucose is converted into two molecules of fructose
2. Which of the following vitamins is required for fatty acid b-oxidation?
 - (a) Vitamin C
 - (b) Vitamin E
 - (c) Vitamin B₁₂
 - (d) Thiamin
 - (e) Riboflavin
3. An example of a discontinuous change over the adult years is the change in
 - (a) Personality
 - (b) Skin elasticity
 - (c) Physical strength
 - (d) Male hormone levels
 - (e) Hair color
4. Deficiency of vitamin A in children causes:
 - (a) Poor muscle development
 - (b) Increased risk of mortality
 - (c) Goitre
 - (d) Increased risk of obesity
 - (e) Increased risk of heart disease
5. Disease can most often be prevented by which of the following:
 - (a) Sunscreen
 - (b) Vaccines
 - (c) Antivirals
 - (d) Pain Relievers
 - (e) Hand Sanitizer

True / False (10 marks)

Answer True or False

6. True or False: Familial hypercholesterolaemia is caused by mutations in the gene that encodes apolipoprotein E.
7. True or False: Phosphocreatine is predominantly found in the cytoplasm of muscle cells, where it plays a key role in energy metabolism.
8. True or False: A stoma should ideally be sited in the rectus sheath abdominus for optimal function and care.
9. True or False: Astroviruses cause widespread replication in the liver and can lead to significant hepatic infection.
10. True or False: Hepatitis C was not first identified through clinical observation but rather through laboratory techniques in the 1980 s.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the limitations of food balance sheets in providing comprehensive insights into food security and nutrition. Support your analysis with relevant examples and critical evaluations.
12. Discuss the effectiveness of various treatments for depression in older adults, and critically analyze why mental stimulation may not be considered an effective intervention.

End of Paper